

TASK -1 ANALYSIS

DATASET: Titanic Dataset

TASK: Understanding Dataset & Data Types

INTRODUCTION :

The aim of this task is to understand the Titanic dataset before applying any machine learning techniques. This analysis helps in deciding whether the dataset is suitable for machine learning and what preprocessing steps might be required.

DATASET OVERVIEW

The Titanic dataset contains **891 rows and 12 columns**.

TYPES OF FEATURES IN THE DATASET

Numerical Features

- Age
- Fare
- SibSp (number of siblings/spouses aboard)
- Parch (number of parents/children aboard)

Categorical Features

- Name
- Sex
- Ticket
- Cabin
- Embarked

Ordinal Feature

- Pclass (Passenger Class)

Binary Feature

- Survived
- Did not survive

TARGET VARIABLE AND INPUT FEATURES

The target variable for this dataset is Survived.

The input features are :

- Pclass
- Sex
- Age
- Fare
- SibSp
- Parch
- Embarked

CATEGORICAL DATA DISTRIBUTION

- The **Sex** column contains male and female passengers, with males being more frequent.
- The **Embarked** column has three categories (C, Q, and S), with most passengers embarking from S.
- The **Pclass** column shows that a large number of passengers belonged to the third class.

DATA QUALITY ISSUES

Missing Values:

- Age
- Cabin
- Embarked

CLASS IMBALANCE

The target variable Survived is slightly imbalanced, with more passengers not surviving compared to those who survived.

CONCLUSION

The dataset size and structure are suitable for basic machine learning tasks but requires addressing the imbalance before proceeding.